

Political Accountability and Local Government Debt: Evidence from China*

Runheng Li¹, Yao Tang^{†1}, Xi Weng¹, and Li-An Zhou¹

¹Department of Applied Economics, Guanghua School of Management, Peking University, 5 Yiheyuan Road, Haidian District, Beijing 100871, China.

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Abstract

This study investigates how the interaction of political accountability and local officials' career incentives shapes the market for Municipal Corporate Bonds (MCBs) in China, taking the 2017 local government debt personal responsibility rule as a quasi-natural experiment. We develop a stylized incomplete-information bargaining model to analyze how the rule reshapes the bargaining equilibrium by rendering officials' observable characteristics credible signals of bailout incentives. Using a dataset of prefecture-level MCBs from 2008 to 2020, we empirically test the model's predictions and focus on separating officials' incentive effects from their inherent ability. Our core findings show that post-announcement of the rule, each additional year of a local party secretary's remaining time to retirement, a proxy for bailout incentives, reduces MCB spreads by approximately 2.5 basis points and increases issuance volume by about 2.0%. These effects are significantly amplified in fiscally stressed cities. Notably, under the 2017 rule, cities led by party secretaries with stronger bailout incentives can expand MCB issuance, which is contrary to the rule's original intent to rein in local borrowing.

JEL classification: H74, G12, E62, P26

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[†]Corresponding author. Email: yao.tang.77@gmail.com

1 Introduction

As China’s local governments seek to finance investment and other economic activities, they have accumulated a massive stock of implicit debt,¹ estimated by the International Monetary Fund at approximately 65.63 trillion yuan (or 9.12 trillion US dollars at current market exchange rates) at the end of 2024.² This large debt burden poses significant risks to both China’s economy and the global economy. Municipal Corporate Bonds (hereafter, MCBs), with an outstanding balance of 15.52 trillion yuan at the end of 2024, constitute a major and visible component of this implicit debt. At the same time, these bonds, issued by Local Government Financing Vehicles (LGFVs), pose a puzzle for understanding China’s public finance and bond markets. Although investors have long recognized the implicit government backing behind MCBs, and no principal default has been recorded to date, these bonds are not priced as de facto sovereign debt. Their issuance yields remain substantially above those of central government bonds and vary significantly across cities and over time. The existing literature on MCBs has focused primarily on pricing from a risk perspective (Liu et al., 2021; Wang et al., 2022; Cao et al., 2025) and, to a lesser extent, on issuance volume (Chen et al., 2020). However, how the debt governance faced by local officials shapes the functioning of the MCB market remains underexplored.

Among a series of debt governance policies that restrict implicit debt taken on by local governments, the personal responsibility rule introduced in 2017 marks a novel policy shift, as it explicitly holds key local officials accountable for debt for the first time.

¹China’s local government implicit debt is generally associated with constrained formal borrowing, continued local spending pressures, and the use of LGFVs and other off-budget vehicles to fund infrastructure (Clarke and Lu, 2017; Chen et al., 2020). Land finance and subnational development incentives further contributed to its expansion (Ang et al., 2018; Gao et al., 2021), while expectations of government support helped sustain this form of quasi-fiscal borrowing (Geng and Qian, 2024).

²Source: International Monetary Fund, *Staff Report for the 2024 Article IV Consultation with the People’s Republic of China*. Webpage link: <https://www.imf.org/en/publications/cr/issues/2024/08/01/people-republic-of-china-2024-article-iv-consultation-press-release-staff-report-and-552803>.

Specifically, this rule, announced at the government’s National Financial Work Conference, imposes lifelong accountability on local party secretaries and mayors for implicit local government debt accrued during their tenure, aimed at curbing reckless borrowing. The management of local government debt has long been plagued by the “soft budget constraint” problem (Kornai, 1986; Kornai et al., 2003), where implicit central government guarantees (Qian and Roland, 1998) and incomplete information (Maskin et al., 2000) weaken local governments’ accountability. As the literature suggests that political accountability can improve fiscal discipline (Besley and Case, 1995; Rodden, 2006; Gordon, 2012), the 2017 personal responsibility rule has the potential to limit the growth in MCB issuance. However, to the best of our knowledge, the impact of the 2017 personal responsibility rule on the MCB market has not been examined.

To fill the gaps, we use a bargaining framework to model how political accountability and local officials’ career incentives shape the MCB market before and after the introduction of the personal responsibility rule, and we test the model’s predictions empirically. In the bargaining framework, local officials and investors are the key players. First, motivated by career incentives (as in Holmström 1999 and Tirole 1994), local officials seek investment-driven growth financed by MCBs, but may have to decide whether to bail out distressed MCBs to avoid paying a personal cost associated with a default. Second, fully aware of the officials’ incentives, investors bargain with officials wanting to issue MCBs. When a local government faces financial stress, the central government provides a fiscal transfer to it with an exogenous probability.

Our theoretical model yields four core propositions that clarify how local officials’ characteristics and city-level fiscal and economic attributes jointly determine the yields and volume of MCBs under the personal responsibility rule. The first and central proposition states that cities led by officials with stronger bailout incentives can issue MCBs at lower yields and in larger volumes, as rational investors demand lower risk premiums and are

willing to accommodate greater issuance. Second, this effect is significantly amplified in cities with a higher probability of fiscal stress. Third, cities that are more dependent on investment-driven growth have weaker bargaining power in the bond market and must offer higher yield spreads. This finding offers a new theoretical rationale for the observation that MCBs sustain positive spreads even in the absence of recorded principal defaults. Lastly, as the personal responsibility rule links bond pricing to heterogeneous official characteristics, variability in MCB yields becomes greater.

We leverage the 2017 personal responsibility rule as a quasi-natural experiment and use a dataset of prefecture-level MCBs spanning 2008-2020 to empirically test the model’s predictions. To disentangle officials’ (specifically party secretaries’) incentive effects from their inherent ability, we employ two complementary identification strategies: including official fixed effects to exploit within-official variation over time, and restricting the sample to bonds issued under party secretaries whose tenure spanned the 2017 policy shift.

Our main empirical findings are as follows. First, consistent with the first proposition, after the introduction of the 2017 rule, each additional year of a local party secretary’s remaining time to retirement, a proxy for stronger bailout incentives, reduces the MCB spread by approximately 2.5 to 3.2 basis points and increases issuance volume by about 2.0%. Second, both the spread-mitigating and volume-expanding effects of party secretaries’ promotion incentives are significantly amplified in cities with higher fiscal stress, supporting Proposition 2. Third, we validate the model’s implied transmission mechanism: party secretaries with stronger promotion incentives raise land transfer income, a main local fiscal tool for mitigating MCB default risks. Lastly, we provide evidence for Proposition 3 using data from Beijing and Shanghai, showing that positive MCB spreads persist in these cities with near-zero default risk. Proposition 4 is also supported, as the standard deviation of MCB yield spreads jumped in 2017.

We further corroborate the model’s propositions with heterogeneity analyses and a

range of robustness checks. Exploiting variation in officials’ career backgrounds, the heterogeneity results provide additional evidence supporting the link between career incentives and MCB yields. The robustness exercises confirm the baseline findings when accounting for the dual incentives of party secretaries and mayors, controlling for the 2014 debt regulation reform, and replacing primary-market spreads with secondary-market counterparts.

The current study contributes to three strands of literature. First, it adds to the literature on the functioning of China’s MCB market. Previous research has suggested that LGFV yields are affected by cities’ fiscal burdens (Liu et al., 2021), local officials’ ability (Wang et al., 2022), the strength of implicit guarantees (Cao et al., 2025), political risk (Ang et al., 2018), and local credit conditions (Huang et al., 2020). On the side of issuance quantity, Chen et al. (2020) shows that the rollover pressure of bank loans obtained by LGFVs during the 2008–2009 stimulus period strongly predicts greater subsequent financing via MCBs. Relative to these studies, we offer a joint analysis of both the pricing and issuance size of MCBs, and we introduce career concerns under political accountability as a previously unexamined determinant of yields and volumes. We are particularly related to Wang et al. (2022) in that we also examine the relationship between local officials’ traits and MCB pricing. Compared to Wang et al. (2022), which focuses on the effect of local officials’ ability on MCB yields, we study how local officials’ incentives influence these yields.

Second, our paper contributes to the literature on career concerns in Chinese federalism. Career concerns are widely recognized as a powerful incentive mechanism in both business and government settings, as they align agents’ behaviors with organizational or societal goals (Holmström, 1999; Tirole, 1994). This theoretical insight finds strong empirical support in the Chinese context: studies show that promotion incentives inherent in China’s political tournament effectively motivate local officials to pursue central growth targets (Maskin et al., 2000; Li and Zhou, 2005; Jia et al., 2015; Xiong, 2018). However,

the incentive role of career concerns is not without limitations. Recent research has highlighted its unintended consequences, such as exacerbating environmental degradation and inducing resource misallocation (Jia, 2017; Persson and Zhuravskaya, 2016). We extend this line of inquiry by identifying a previously unexplored, consequential limitation: the personal responsibility system, designed to curb reckless borrowing, paradoxically enables more debt accumulation in cities led by officials with strong promotion prospects. Our key insight is that capital markets treat an official’s reputation and career trajectory as “political collateral”, boosting credit supply to those most averse to default. Thus, political incentives, when paired with market pricing, can partially undermine the central government’s debt-stabilization goals.

Lastly, this work relates to a literature on political accountability and fiscal discipline. The theory of political selection posits that accountability mechanisms, such as term limits or personal responsibility rules, should in principle discipline fiscal policy and reduce moral hazard (Besley and Case, 1995; Besley and Burgess, 2002). In the context of “soft budget constraints” (Kornai, 1986; Poterba, 1994; Qian and Roland, 1998), the imposition of lifelong responsibility is considered as a solution to local fiscal irresponsibility. However, our study nuances this conventional wisdom by showing that accountability rules’ effectiveness depends on market perceptions of these constraints. While accountability raises an agent’s default costs, in a marketized debt setting, it also acts as a selection mechanism granting high-incentive officials better access to capital. Once accounting for officials’ career concerns, personal accountability may not guarantee fiscal discipline.

The rest of this paper proceeds as follows. In section 2, we construct a sequential game under incomplete information to illustrate how the personal responsibility rule shapes officials’ incentives, investor behavior, and MCB market outcomes. We present the empirical specification and data description in section 3, and report the main empirical results in section 4. In section 5, we conduct heterogeneity analysis and robustness tests.

Finally, we conclude the paper in section 6.

2 Theory

We analyze the functioning of the market for bonds issued by an LGFV in a sequential game with incomplete information. The players in the game are the local government, the LGFV, a representative investor and the central government. To simplify the analysis, we assume that both the local government and the LGFV are led by the local official. Our main interests are the determination of issuance size and interest rate of the MCBs.

2.1 Baseline Model

The local government official wants to issue a bond via the LGFV. The time line of the issuance and repayment of MCBs is as follows:

- i. Issuance phase. The local official and the investor determine the issuance size x and the interest rate r through bargaining.
- ii. Interim phase.
 - (a) The LGFV faces financial stress with a city-specific probability p , exposing its bond to distress risk.
 - (b) If financial stress occurs, the local official decides whether to raise additional funds to bail out the distressed MCBs.
 - (c) If the local official decides not to bail out the MCB, the central government may provide a fiscal transfer to the LGFV with an exogenous probability λ to prevent a principal default.
- iii. Repayment phase. If financial stress occurs and neither the local official nor the central government intervenes, the LGFV defaults, causing the investor to lose the principal and receive no interest. If the local official does not intervene but the central government makes a transfer, the investor recovers the principal but receives

no interest. Otherwise (i.e., if no stress occurs or the local official bails out the distressed bond), the investor receives full payment of both the principal and the interest.

When a bond is issued successfully to fund investment in the local economy, the economic output of the local economy increases by $y(x)$, where $y(x)$ is a strictly concave function. In the event of a default, irrespective of whether the central government intervenes, the local government official is subject to a loss ω per unit of bond. To avoid default after the occurrence of financial stress, the local official can raise additional funds to bail out the distressed bond. The fund-raising requires the official to pay a constant effort of c per unit of debt. We denote the decision to implement a bailout with an indicator function, $b(\omega)$, as the decision hinges on the comparison between the personal loss ω associated with default and the effort cost c associated with fundraising:

$$b(\omega) = \mathbb{1} (\omega > c). \quad (1)$$

The extensive game tree is presented in Figure 1.

In the baseline model, the personal loss that a local official incurs from bond defaults is equal to a random variable ε :

$$\omega = \varepsilon. \quad (2)$$

The random variable ε is assumed to be uniformly distributed over the interval $[0,1]$, which implies that the cdf function is $F(\varepsilon) = \varepsilon$. At the time of bargaining, both the local official and the investor know the cumulative distribution function. Subsequently, the official learns about the realization of ε in the interim phase between the time the bond is successfully issued and the time for repayment. The idea is that after the issuance of an MCB, random factors, such as changes in the inclination of central government to punish local governments for defaults and availability of promotional opportunities, can affect personal loss associated with a default.

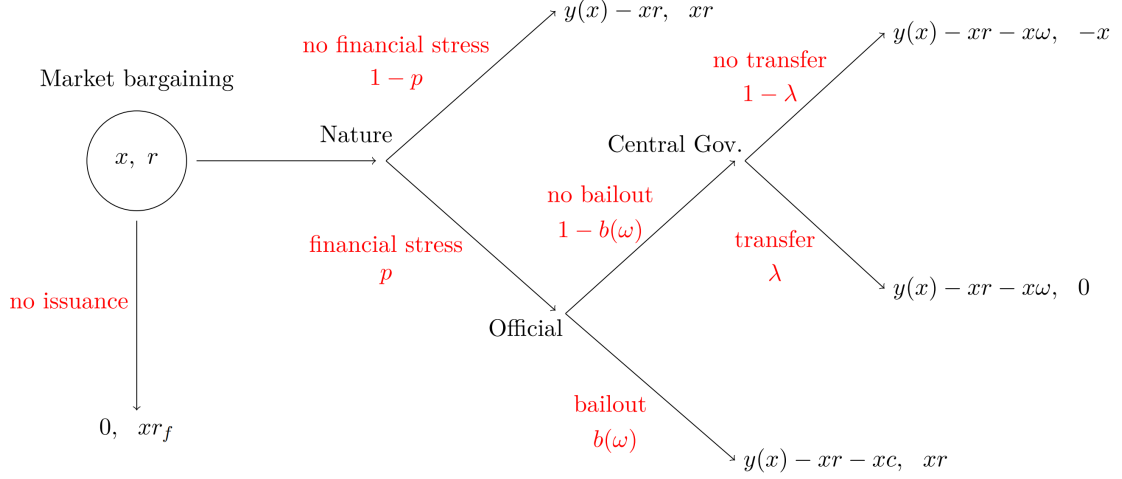


Figure 1. Game of Issuance of Municipal Corporate Bonds

Notes: The probability settings at the “Nature” node essentially reflect expectation divergence under information asymmetry: officials know their bailout incentives, while investors infer risks inversely through observable characteristics. This information structure is the core premise for the rule to reshape the bargaining equilibrium.

Conditional on ε , the expected payoff for the local official is

$$\begin{aligned}
 U_L(x, r, p \mid \varepsilon) &= (1 - p)(y(x) - xr) + pb(\omega)(y(x) - xr - xc) + p(1 - b(\omega))(y(x) - xr - xw) \\
 &= y(x) - xr - \min(\varepsilon, c)px.
 \end{aligned} \tag{3}$$

At the bargaining phase when ε is unknown, the expected payoff to the local official is

$$U_L(x, r, p) = \mathbb{E}_\varepsilon[U_L(x, r, p \mid \varepsilon)] = y(x) - xr - \hat{c}px, \tag{4}$$

where

$$\hat{c} = \mathbb{E}_\varepsilon[\min(\varepsilon, c)] = \int_{-\infty}^c \varepsilon dF(\varepsilon) + \int_c^{+\infty} c dF(\varepsilon), \tag{5}$$

can be interpreted as an adjusted bailout cost.

From the investor’s point of view, the expected value of $b(\omega)$ is

$$\mathbb{E}[b(\omega)] = \text{Prob}(\varepsilon > c) = 1 - F(c). \tag{6}$$

If the investor buys the bonds, the expected payoff at the bargaining phase is

$$\begin{aligned} U_I(x, r, p) &= (1 - p)xr + p\mathbb{E}[b(\omega)]xr - p(1 - \mathbb{E}[b(\omega)])(1 - \lambda)x \\ &= xr - pF(c)(r + 1 - \lambda)x. \end{aligned} \tag{7}$$

2.2 Model with Personal Responsibility

In the baseline model, if an MCB is defaulted, the local official faces a random loss that is not person-specific; for instance, a rebuke of all top-ranking officials in a city acting as a decision-making team. This setup is broadly consistent with reality up to 2017, when a personal responsibility rule was introduced by the National Financial Work Conference in the year. The rule represents a major change in the regulation of local government debt, as it explicitly states that top local officials would remain accountable for debt issued during their tenure for their lifetime.³

To capture the effect of the personal responsibility rule, in this subsection, we modify the personal loss so that it is equal to the sum of the loss associated with punishment ζ and the random exogenous factor ε :

$$\omega = \zeta(z) + \varepsilon. \tag{8}$$

where punishment-related loss ζ depends on z , the personal characteristic of an official. Without loss of generality, we assume that ζ is strictly increasing in z . An important element of z in China is the number of years remaining before an official reaches retirement age: if a younger official receives a punishment and thus is significantly held back in his or her career path, the punishment-related loss is larger for the official. The higher the value of z , the stronger the incentive to avoid the default of an MCB.

Similarly to equation (4), the expected payoff to the local official before the debt

³Explicit local government debt issued in accordance with the approval of the National People's Congress is not subject to the rule. LGFV debt is typically not part of explicit government debt and is hence subject to the regulation of the new rule.

issuance can be written as

$$U_L(x, r, z, p) = \mathbb{E}_\varepsilon[U_L(x, r, z, p \mid \varepsilon)] = y(x) - xr - \hat{c}(z)px, \quad (9)$$

where the adjusted bailout cost is

$$\hat{c}(z) = \mathbb{E}_\varepsilon[\min(\zeta(z) + \varepsilon, c)] = \int_{-\infty}^{c-\zeta(z)} (\zeta(z) + \varepsilon) dF(\varepsilon) + \int_{c-\zeta(z)}^{+\infty} c dF(\varepsilon), \quad (10)$$

which is therefore related to personal characteristic z .

Similarly to equation (7), the expected payoff to the investor of buying the bond is

$$U_I(x, r, z, p) = xr - pF(c - \zeta(z))(r + 1 - \lambda)x. \quad (11)$$

2.3 Bargaining Solution

In line with the book-building issuance method⁴ used in practice, we adopt a bargaining framework in which the issuer and the investor determine the interest rate and issuance volume through Nash bargaining. The LGFV issuer possesses a relative bargaining power of $\theta \in (0, 1)$. If the bond offering is unsuccessful, the investor will obtain a default yield of r_f , which can be considered as the risk-free return on the market. The local official will receive a zero payoff.

The maximization problem is

$$\max_{x, r} [U_L]^\theta [U_I - xr_f]^{1-\theta}. \quad (12)$$

Substituting equations (4) and (7) into the first order conditions of the last line, we can obtain **the bargaining solution in the baseline model**:

$$y'(x) = \frac{r_f + pF(c)(1 - \lambda)}{1 - pF(c)} + \hat{c}p, \quad (13)$$

$$r = \frac{r_f + pF(c)(1 - \lambda)}{1 - pF(c)} + (1 - \theta)\left(\frac{y(x)}{x} - y'(x)\right). \quad (14)$$

⁴China's MCBs are typically issued through a book-building process. In the process, the lead underwriter, acting as the bookrunner, determines an initial interest rate range via market inquiry, records investors' subscription orders, and then negotiates the final price with the issuer before completing the allocation.

Substituting equations (9) and (11) into the maximization problem, **the bargaining solution under the personal responsibility rule** becomes

$$y'(x) = \frac{r_f + pF(c - \zeta(z))(1 - \lambda)}{1 - pF(c - \zeta(z))} + \hat{c}(z)p, \quad (15)$$

$$r = \frac{r_f + pF(c - \zeta(z))(1 - \lambda)}{1 - pF(c - \zeta(z))} + (1 - \theta)\left(\frac{y(x)}{x} - y'(x)\right). \quad (16)$$

Equation (15) implicitly defines the issuance volume as a function of the exogenous variables and parameters. After obtaining the solution for volume, $x^*(z, p)$, we can subsequently solve for the equilibrium interest rate $r^*(z, p)$ from equation (16). The strict concavity of $y(x)$ ensures that $y'(x)$ is a monotonically decreasing function with a range of $(0, +\infty)$, which guarantees the existence and uniqueness of the bargaining solution.

Compared to the bargaining solution in the baseline case, the bargaining solution under the personal responsibility rule features an equilibrium issuance interest rate and volume that are determined not only by the fundamentals of a city (probability of financial stress p and the production technology embodied in $y(x)$), but also by the loss-related individual characteristics of the local official z . The difference is important: post the introduction of the personal responsibility rule, the bargaining equilibrium changes from a pooling equilibrium to a separating equilibrium as the market begins to discern differences in local officials.

2.4 Propositions

In this subsection, we present four propositions related to the comparative statics with respect to local officials' characteristics and city-level fiscal risks, and the composition and variability of yield spreads. Proofs of the propositions can be found in the Appendix.

Proposition 1. *After the introduction of the personal responsibility rule, (a) the yield of an MCB is a decreasing function of z , the characteristic of an official that is positively related to the incentive to bail out a distressed bond, and (b) the issuance volume of an MCB is an increasing function of z .*

The core mechanism behind Proposition 1 is that the personal responsibility rule shapes the behavior of both the local official and the investor. Following the introduction of the new rule, officials with a characteristic that is related to a larger default-related loss (such as the number of years remaining before an official reaches retirement age) will endogenously make a greater effort to bail out the distressed MCB. The investor who understands the incentive will therefore accept a lower risk premium for the MCBs issued under the direction of such local officials and accommodate a larger issuance size. The introduction of personal responsibility makes the observable characteristics of local officials relevant to the bargaining process.

Meanwhile, the direction of change in the aggregate level of bond issuance is ambiguous. If more officials can guide LGFVs under their leadership to larger bond issues, then the total debt level of local governments will increase. Notably, such a potential equilibrium outcome would contradict the intent of the personal responsibility rule, which is to control the debt level of local governments.

Proposition 2. *Following the introduction of the personal responsibility rule, the impact of a local official's characteristic on yields and issuance volume of an MCB is more pronounced in cities with a higher probability of fiscal stress.*

Intuitively, in cities with a higher chance of fiscal stress, the uncertainty about the repayment of MCBs is inherently greater, and investor concerns about bond defaults are more prominent. In such cities, officials with characteristics that induce a greater bailout effort are valued more by the investors, leading to both a lower risk premium and a larger issuance size in equilibrium. Meanwhile, for cities with sound fiscal conditions and hence a low risk of default, the reassuring effect of officials' characteristics would only have a negligible effect on the bargaining outcome.

While Propositions 1 and 2 focus on how MCB issuance responds to officials' incentives and dynamic fiscal risks, our bargaining framework also illuminates the structural

nature of MCB pricing. In the next proposition, we show that beyond city-specific financial risks and official characteristics, a city's reliance on investment-driven growth shapes the equilibrium by weakening the LGFV's bargaining power vis-à-vis investors.

Proposition 3. *The equilibrium yield spread of MCBs consists of two components:*

1. A **risk premium**, which compensates investors for the probability of default; and
2. An **investment-dependence premium**, which reflects the local government's weakened bargaining power stemming from its reliance on debt-fueled economic growth.

This general decomposition follows directly from the bargaining solution in Equation (16), separating the baseline risk-adjusted rate from the bargaining wedge:

$$r^*(z, p) = \underbrace{\frac{r_f + pF(c - \zeta(z))(1 - \lambda)}{1 - pF(c - \zeta(z))}}_{\text{Risk-adjusted benchmark rate}} + \underbrace{(1 - \theta) \left(\frac{y(x^*)}{x^*} - y'(x^*) \right)}_{\text{Investment-dependence premium}}. \quad (17)$$

To highlight the importance of this investment-driven dependence in the pricing of MCBs, we consider a limiting case in which the fundamental risk of municipal financial stress vanishes.

Corollary 1. *In a limiting scenario where the probability of municipal financial stress and hence the risk premium approach zero, MCBs still exhibit a strictly positive yield spread over the risk-free rate, which is driven entirely by the investment-dependence premium.*

Taking the limit $p \rightarrow 0$ in the equilibrium equation, the risk-adjusted benchmark rate converges cleanly to the risk-free rate r_f , isolating the bargaining wedge:

$$\lim_{p \rightarrow 0} r^* = r_f + (1 - \theta) \left(\frac{y(x^*)}{x^*} - y'(x^*) \right) > r_f. \quad (18)$$

The mathematical foundation for this strictly positive premium lies in the strictly concave production function $y(\cdot)$, which ensures that the average product of public investment always exceeds its marginal product (i.e., $\frac{y(x^*)}{x^*} > y'(x^*)$). Driven by career incentives

to maximize local economic output, local officials must secure sufficient capital from the market. To achieve this, they are compelled to cede a portion of the economic surplus to investors, generating a premium proportional to the investor’s bargaining power $(1 - \theta)$.

By isolating this investment-dependence premium, Corollary 1 suggests a novel explanation for the long-standing puzzle in the Chinese bond market: why MCBs consistently trade at spreads above sovereign benchmarks despite the historical absence of principal defaults. The model suggests that persistent positive spreads reflect more than default risk, as part of the spreads reflects local officials’ reliance on debt-driven growth.

After establishing how official incentives and structural growth dependence shape equilibrium pricing, we turn to the cross-sectional heterogeneity induced by the 2017 reform.

Proposition 4. *The introduction of the personal responsibility rule leads to greater variability in the MCB interest rates between cities.*

Intuitively, the introduction of the personal responsibility rule enables investors to assess the likelihood of a bailout by local officials based on their observable characteristics. Officials with characteristics that make them less sensitive to punishment at default are compelled to offer higher interest rates on the bonds they issue, as they are less likely to bail out the MCB during financial stress. As personal characteristics become priced in equilibrium post the personal responsibility rule, variation in characteristics leads to greater dispersion in bargaining outcome.

3 Empirical Specification and Data

3.1 Regression Models

We estimate two regression models to test the comparative statics of Proposition 1 and Proposition 2. **The first regression model** examines the relationship between the

characteristics of a city’s party secretary and the yield spread of MCBs issued by the city.

$$\begin{aligned} Spread_{ict} = & \beta_0 + \beta_1 \cdot z_{ct} + \beta_2 \cdot Policy_t + \beta_3 \cdot z_{ct} \cdot Policy_t \\ & + \gamma \cdot Controls + \psi_c + \iota_o + \eta_t + \epsilon_{ict}. \end{aligned} \quad (19)$$

The dependent variable, $Spread_{ict}$, is the difference between the issuance interest rate of MCB i issued by city c in month t and the risk-free interest rate of the same maturity on the issuance date. It measures the riskiness of the bond as perceived by investors.

The key independent variable z_{ct} is *time to retirement*, a key characteristic of the incumbent party secretary in city c in time t that is positively associated with the personal loss from bond distress. Time to retirement is measured as the years remaining until the party secretary reaches the statutory retirement age.⁵ This variable is strongly correlated with the career prospects of the party secretary, the top local official in the city, since age ceilings constitute a rigid constraint on their career advancement. Since younger officials have more opportunities for advancement in the government hierarchy, this variable is commonly used to measure an official’s career incentives (Li and Zhou, 2005; Bertrand et al., 2020). As officials approach the mandatory retirement age, their promotion prospects vanish, and career concerns weaken, leading to behavioral shifts (Persson and Zhuravskaya, 2016; Xiong, 2018).

The change in the incentive regime is measured by $Policy_t$, a binary variable that takes a value of 0 before the introduction of the personal responsibility rule in July 2017 and 1 thereafter. A negative estimated value of β_3 , the coefficient on the interaction between $Policy_t$ and z_{ct} , would support the key prediction of Proposition 1a: after the introduction of personal responsibility, cities with younger party secretaries will be able to sell bonds at a lower risk premium.

Control variables include the maturity of bonds, the logarithmic of GDP, and the logarithmic of GDP per capita. ψ_c , ι_o , η_t , and ϵ_{ict} denote the city fixed effects, official

⁵In China, the statutory retirement age is generally 60 for prefecture-level officials and 63 for sub-provincial officials. We account for this institutional difference when constructing the measure of time to retirement.

fixed effects, year fixed effects, and error term, respectively.

The second regression model examines whether the effect of party secretaries' career incentives on MCB yield spreads is stronger in cities facing greater fiscal stress. To be specific, we introduce a city-level proxy variable, $FinStress_{ct}$, defined as fiscal revenue divided by GDP in city c and time t . Since a lower ratio indicates weaker fiscal capacity, this variable is an inverse measure of fiscal stress. We then interact $FinStress_{ct}$ with both the personal responsibility rule and the party secretary's time to retirement. The estimating equation is

$$\begin{aligned}
Spread_{ict} = & \phi_0 + \phi_1 \cdot z_{ct} + \phi_2 \cdot Policy_t + \phi_3 \cdot FinStress_{ct} \\
& + \phi_4 \cdot z_{ct} \cdot Policy_t + \phi_5 \cdot z_{ct} \cdot FinStress_{ct} + \phi_6 \cdot Policy_t \cdot FinStress_{ct} \\
& + \phi_7 \cdot z_{ct} \cdot Policy_t \cdot FinStress_{ct} \\
& + \mu \cdot Controls + \psi_c + \iota_o + \eta_t + \epsilon_{ict}.
\end{aligned} \tag{20}$$

The model's primary focus is the triple interaction term. A positive estimated value of ϕ_7 would support the prediction of Proposition 2: after the introduction of the personal responsibility rule, the spread-reducing effect of a longer time to retirement is stronger in fiscally more stressed cities.

In addition to yield spreads, the model generates a parallel set of predictions for MCB issuance volume (Propositions 1b and 2b). In particular, party secretaries' promotion incentives should affect issuance volume with the opposite sign to their effect on spreads. We test these implications by re-estimating the two specifications above with issuance volume as the dependent variable.

3.2 Data

Our empirical analysis combines five data sources. MCB data, including issuance yield, size, maturity, and other characteristics, are obtained from the *Wind* database. Official data, including age, tenure, career background, and other characteristics, are compiled by Cao et al. (2019). City-level economic variables are retrieved from the *China City*

Statistical Yearbook. Land transfer data, used in mechanism analysis, are sourced from the *China Land Market Network*. ChinaBond valuation yields, used in robustness checks, are from the *ChinaBond Valuation Center*.

The merged sample contains 24,474 bond-official-city observations from January 2008 to December 2020. To isolate the effect of career incentives from officials' time-invariant inherent ability, we additionally construct a restricted sample comprising bonds issued under party secretaries whose tenure spans the July 2017 policy shift. Table 1 reports summary statistics.

To analyze the temporal and spatial pattern of the volume of MCBs before and after the policy change in July 2017, in Figure 2 we present the maps of the annual MCBs to GDP ratios in prefecture-level administrative units in China in 2015 and 2020. During the five-year period, the overall issuance scale expanded substantially, reflecting a greater reliance on MCB financing. For the intensive margin, cities that had issued MCBs by 2015 recorded an average increase of 1.46% in the ratio of their annual bond issuance volume to GDP between 2015 and 2020. For the extensive margin, the number of cities with zero cumulative MCB issuance declined from 114 to 72 by 2020, indicating a broader participation of cities in the MCB market. The expansion of MCBs on both margins reveals that the personal responsibility rule has been unsuccessful in stopping the growth of MCBs.

Figure 3 presents the pattern of MCB issuance volumes over time. The left panel tracks the volume of bonding at the median, the 60th, the 70th, and the 80th percentiles at the city level for each year. The right panel displays the annual aggregate issuance volume in four major regions (Eastern China, Central China, Western China, and Northeast China). From both panels, it is obvious that, following the introduction of the personal responsibility rule, bond issuance volumes generally rose. As discussed in the theory section, if more officials are able to guide LGFVs under their leadership to larger bond

issues after the policy change, the total debt level of local governments will increase.

4 Main Empirical Results

This section presents empirical tests of the model’s key predictions. We begin by examining how the 2017 personal responsibility rule reshapes the relationship between local officials’ career incentives and MCB yield spreads, followed by a parallel analysis of issuance volumes. We then investigate the mechanism through which officials with stronger promotion incentives reduce default risk, and provide evidence on the investment-dependence premium and increased yield dispersion.

A central identification challenge is distinguishing the incentive channel from ability-based explanations. Officials with longer time to retirement may differ systematically in administrative competence or political connections — characteristics that could independently affect MCB pricing. To isolate the incentive channel, we employ two strategies. First, in the full-sample regressions, we include official fixed effects, which absorb all time-invariant characteristics of the party secretary. Since time to retirement declines mechanically as an official ages during their tenure, the interaction term $TTR \times Policy$ is identified from within-official variation over time, differencing out fixed ability. Second, the restricted-sample regressions provide a sharper test by focusing exclusively on bonds issued under party secretaries whose tenure spans the July 2017 policy change. This design eliminates concerns that officials with different career horizons are systematically assigned to cities with different fiscal conditions, a form of endogenous matching that could confound cross-official comparisons. Since we observe the same official before and after the policy shock, the interaction between pre-existing career horizons and the post-2017 indicator isolates how the policy differentially activates career incentives, a source of variation orthogonal to ability by construction.

4.1 Career Incentives and MCB Yield Spreads

This subsection tests the predictions of Proposition 1a and Proposition 2, which link MCB issuance spreads to officials' career incentives under the 2017 personal responsibility rule and to city-level fiscal conditions.

Table 2 first provides evidence on Proposition 1a. Column (1), based on equation (19), shows that the interaction between the party secretary's time to retirement and the personal responsibility rule is negative and statistically significant at the 1% level. The point estimate of -3.231 indicates that the 2017 rule meaningfully shifted how investors price officials' career horizons. Each additional year of the party secretary's remaining time to retirement is associated with a 3.231-basis-point reduction in MCB spreads in the post-reform period relative to the pre-reform period. This result is consistent with the model's prediction that personal accountability makes officials' observable characteristics credible signals of bailout incentives to investors.

Column (2) tests Proposition 2, which predicts that this policy-induced shift should be stronger in cities facing greater fiscal stress. We proxy fiscal stress inversely using the ratio of fiscal revenue to GDP, so a lower value indicates weaker fiscal capacity and higher fiscal risk. The triple interaction term is positive and statistically significant at the 1% level, with a coefficient of 1.185. This suggests that cities with weaker fiscal capacity exhibit a larger policy-induced shift in how spreads respond to TTR, with each 1-percentage-point reduction in the fiscal revenue ratio associated with an additional 1.185-basis-point larger reduction in spreads per year of remaining tenure.

Columns (3) and (4) employ the restricted sample, which confines attention to bonds issued under party secretaries whose tenures span the 2017 policy change. In Column (3), the coefficient on the interaction term is -2.548 and remains statistically significant, close in magnitude to the full-sample estimate. Based on this estimate, a one-standard-deviation increase in time to retirement (3.117 years), is predicted to cause a 7.942-basis-point

reduction in spreads, confirming that the policy-induced shift is economically meaningful. Column (4) applies the same restricted-sample strategy to test Proposition 2. The triple interaction term is positive, statistically significant at the 1% level, and larger in magnitude (1.538) than in the full sample.

The consistency of results across both the full and restricted samples confirms that the estimated effects capture the policy’s causal impact rather than endogenous sorting of officials across cities. By imposing personal accountability, the 2017 rule significantly amplified the role of party secretaries’ promotion incentives as a priced factor in bond yields, with this effect particularly pronounced in fiscally vulnerable cities.

4.2 Career Incentives and MCB Issuance Volumes

We next turn to the volume of bonds. Proposition 1b predicts that, following the introduction of the 2017 personal responsibility rule, cities led by party secretaries with stronger career incentives should issue more MCBs. Proposition 2 further implies that this effect should be stronger in cities facing greater fiscal stress. Table 3 reports the corresponding estimates, using log issuance volume as the dependent variable.

Columns (1) and (2) present the full-sample results. Column (1) shows that the interaction between the party secretary’s time to retirement and the post-2017 policy dummy is positive and statistically significant at the 1% level. The coefficient of 0.0199 indicates that, after the reform, each additional year of the party secretary’s remaining time to retirement is associated with about a 1.99% increase in MCB issuance volume. This result is consistent with Proposition 1b and suggests that once personal accountability makes officials’ bailout incentives more salient to investors, cities led by younger party secretaries are able to place more debt. Column (2) tests Proposition 2 by introducing the interaction with fiscal stress. The triple interaction term is negative and statistically significant at the 5% level. Since fiscal revenue as a share of GDP is an inverse measure of fiscal stress, this estimate implies that each 1-percentage-point reduction in the fiscal

revenue ratio is associated with an additional 0.587% increase in issuance volume per year of remaining tenure after the policy.

Columns (3) and (4) repeat the analysis in the restricted sample of bonds issued under party secretaries whose tenure spans the 2017 policy change. In Column (3), the coefficient on the interaction term remains positive and statistically significant at the 1% level, at 0.0200, nearly identical to the full-sample estimate. Each additional year of the party secretary’s remaining time to retirement is associated with about a 2.00% increase in issuance volume in the post-reform period. Evaluated at the restricted-sample mean issuance size, this corresponds to an increase of about 17.6 million yuan per bond. Column (4) shows that this effect is again significantly stronger in cities with weaker fiscal capacity. The triple interaction term is negative and statistically significant at the 5% level, implying that each 1-percentage-point reduction in the fiscal revenue ratio is associated with a further 0.650% increase in issuance volume per year of remaining tenure.

Taken together, the evidence on issuance volumes closely parallels the pricing results in Table 2. After the introduction of the personal responsibility rule, stronger promotion incentives not only lower MCB spreads but also enable cities to issue more debt, and both effects are more pronounced in fiscally stressed cities.

4.3 Land Transfers as a Risk-Mitigation Channel

In theory, party secretaries with stronger career incentives are more willing to mobilize fiscal resources to prevent bond distress. In the Chinese institutional context, land transfer is one of the most important fiscal tools available to local officials (Bai et al., 2016; Ang et al., 2018). As a core source of revenue in China’s local fiscal system, land transfer income often serves as a buffer for local debt repayment.⁶ If the 2017 personal responsibility rule raises the personal cost of bond distress, then politically ambitious party

⁶In 2021, the peak of China’s real estate market, land transfer income accounted for 23.4% of total revenue across all levels of government in China. By 2024, after the real estate downturn, the ratio had fallen to 12.0%.

secretaries should have stronger incentives to expand land-based fiscal resources. We therefore examine whether cities led by party secretaries with longer time to retirement manage to obtain more land transfer revenue after the policy change.

Table 4 reports the results. Columns (1) and (2) use the full city-month sample. In Column (1), where monthly land transfer revenue is the dependent variable, the interaction between the party secretary's time to retirement and the post-2017 policy indicator is positive and statistically significant at the 5% level. The coefficient implies that, after the introduction of the personal responsibility rule, each additional year of remaining time to retirement is associated with an increase of about 56.0 million yuan in monthly land transfer revenue. Column (2) shows a similar pattern when the dependent variable is land transfer revenue per capita. The coefficient on the same interaction term is statistically significant at the 1% level, implying an increase of about 11.14 yuan in monthly land transfer revenue per capita for each additional year of remaining tenure after the policy.

Columns (3) and (4) utilize the restricted city-month sample, which includes only observations associated with party secretaries whose tenure spans the 2017 policy change. The estimated interaction terms remain positive and statistically significant in both specifications, consistent with the full-sample results.

Taken together, the evidence suggests that the personal responsibility rule strengthens the incentive of politically ambitious party secretaries to mobilize land finance. Cities led by party secretaries with longer time to retirement experience larger post-2017 increases in both aggregate and per-capita land transfer revenue, providing them with greater fiscal resources to mitigate bond repayment risk. These results support the mechanism emphasized in the model: officials with stronger incentives will make more efforts to avoid default.

4.4 Investment Dependence and the Persistence of Positive Spreads

Proposition 3 decomposes the MCB yield spread into two components: a standard risk premium and an investment-dependence premium. Directly testing this decomposition in the data is difficult, because for typical prefecture-level cities credit risk and investment dependence are closely intertwined. As a result, standard observational variation is unlikely to isolate the investment-dependence premium cleanly from the risk premium.

We therefore turn to an empirically tractable implication of the limiting case in Corollary 1. As the probability of severe fiscal stress becomes negligible ($p \rightarrow 0$), the risk premium is largely eliminated, so the remaining spread should primarily reflect the investment-dependence premium. Under typical asset-pricing logic, the spread of an issuer with near-zero default risk should converge to zero. By contrast, our framework predicts that the spread can remain persistently positive because local governments rely on debt financing to support growth.

To examine this implication, we study MCBs issued by Beijing and Shanghai. As China’s political and economic centers, these two municipalities have exceptionally strong fiscal capacity.⁷ Relative to typical local governments, their default risk is plausibly close to negligible, making them useful empirical proxies for the boundary case in Corollary 1. Figure 4 plots the average issuance yields of 1-year, 3-year, and 5-year MCBs in Beijing and Shanghai together with the corresponding risk-free rates, measured by central government bond yields of the same maturity. Across all maturities, MCB yields remain substantially above the risk-free benchmark. Indeed, the risk-free rate lies below the one-standard-deviation lower bound of MCB yields in both municipalities.

These patterns are consistent with the implication of Corollary 1 and, more broadly, with the decomposition in Proposition 3. Even when default risk is minimal, local govern-

⁷According to data from the National Bureau of Statistics of China, Beijing’s and Shanghai’s general public budget revenue as a share of GDP averaged 17.7% and 19.7% over 2008–2020, respectively, substantially higher than the sample mean of 7.3% reported in Table 1.

ments still pay a positive spread because their dependence on debt-financed investment weakens their bargaining position vis-à-vis investors.

4.5 Increased Yield Dispersion

Proposition 4 predicts that the introduction of the personal responsibility rule should increase the cross-city dispersion of MCB yields. Figure 5 provides descriptive evidence consistent with this prediction. It plots the time trends of two city-level pricing measures for MCBs, coupon rates and spreads over the risk-free rate, for 1-year and 5-year bonds. Visually, the cross-city range of these pricing measures widens after the introduction of the personal responsibility rule, especially for shorter-maturity bonds.

To test Proposition 4 more formally, we implement a time-discontinuity regression that exploits the introduction of the personal responsibility rule in July 2017, with time measured in months relative to the policy cutoff. The dependent variable is the monthly cross-city standard deviation of MCB yields, and the specification allows for different time trends on both sides of the cutoff. Table 5 reports the results. For 1-year bonds, we find a statistically significant upward jump in the cross-city dispersion of both coupon rates and yield spreads at the policy cutoff. For 5-year bonds, the estimated discontinuities are statistically insignificant. Figure 6 visualizes the regression outcome and shows a clear increase in cross-city dispersion around the policy threshold for short-maturity bonds.

The difference in spread variability between 1-year and 5-year bonds is consistent with the mechanism of the model. For longer-maturity bonds, repayment is more likely to occur after the issuing official has left office and hence depends on fiscal decisions made by the successor. Because the successor does not bear the same personal responsibility for debt issued during the predecessor’s tenure, his incentives to allocate fiscal resources to such debt are generally weaker than for debt issued during his own tenure. As a result, the issuing official’s observable characteristics are a more informative signal of expected repayment support for shorter-maturity bonds than for longer-maturity bonds. Overall,

the evidence supports Proposition 4: by making officials' heterogeneous incentives more relevant for pricing, the 2017 personal responsibility rule increased cross-city dispersion in MCB yields.

5 Heterogeneity Analysis and Robustness Tests

5.1 Heterogeneity in Officials' Career Background

In this subsection, we exploit heterogeneity in officials' career backgrounds to provide further evidence on the mechanism underlying the personal responsibility rule. The model predicts that, after 2017, investors use observable career characteristics to infer bailout probabilities and price MCBs accordingly. A natural implication is that investors should price officials' career characteristics more strongly when those characteristics are more predictive of actual repayment support and when promotion incentives are more closely connected to local bond rescue.

We consider two dimensions of career background. The first is whether the party secretary was promoted from the mayoralty of the same city. We distinguish between mayor-turned secretaries, who were promoted locally, and other secretaries, who were transferred from elsewhere. This dimension captures local institutional embeddedness. Bailout incentives matter only if officials can translate them into actual intervention. In practice, this may involve coordinating local banks and SOEs, or mobilizing fiscal and administrative resources to support repayment. Secretaries promoted from within the same city are more likely to have the local knowledge and working relationships needed to carry out such coordination, making their bailout incentives more likely to result in actual repayment support.

The second dimension is whether the party secretary has prior central-government experience. While a longer time to retirement generally implies better promotion prospects, for secretaries with central-government experience, future advancement may not depend

solely on local performance. Their career prospects may also hinge on broader bureaucratic networks and career paths outside the locality. This weakens the link between local default risk and personal career loss. Consequently, compared with officials without central experience, this group’s promotion incentives may be less tightly coupled to local bond rescue.

Table 6 reports the results. Columns (1)–(4) use the full sample; Columns (5)–(8) restrict to bonds issued under secretaries whose tenures span the 2017 policy shift. Among mayor-turned secretaries, the interaction coefficient is negative and significant (−3.686 in Column 1, −2.881 in Column 5); for other secretaries, it is smaller and insignificant (−2.197 in Column 2, 0.593 in Column 6). Similarly, for secretaries without central experience, the coefficient is significantly negative (−3.881 in Column 4; −3.480 in Column 8); for those with central experience, it is small and insignificant (−2.742 in Column 3; −0.884 in Column 7).

Overall, the spread-reducing effect of career incentives is concentrated among mayor-turned secretaries and among officials without central-government experience. These patterns support the model’s mechanism: local embeddedness determines whether bailout incentives can be operationalized, while central experience attenuates the channel through which local default risk feeds into personal career risk, and hence into rescue incentives. Taken together, these results suggest that, after the personal responsibility rule, investors price officials’ career incentives more strongly when those incentives are both more actionable and more tightly linked to local debt outcomes.

5.2 Robustness Tests

This subsection further assesses the robustness of the baseline results by considering mayors (another types of local officials), an earlier policy shock, and an alternative measure of bond spreads.

First, we examine whether MCB pricing mainly responds to the characteristics of

the party secretary or those of the mayor. Although the 2017 personal responsibility rule applies to both party secretaries and mayors, a party secretary, as the first-in-command in cities in China, typically has greater authority than the mayor over local fiscal strategy and land resources. Table 7 replaces the secretary-based retirement measure with the mayor's counterpart and then includes both simultaneously. The interaction term based on the mayor is insignificant, while the secretary-based interaction remains significantly negative when both are included. The same pattern holds in the restricted sample that requires both officials' tenures to span the 2017 policy change. These results confirm that the MCB market mainly prices the career incentives of party secretaries rather than those of mayors.

Second, we account for the earlier debt-policy tightening introduced in 2014 to ensure that the baseline results are not simply capturing the effect of the earlier debt regulation. The 2014 policy titled *Opinions of the State Council on Strengthening the Management of Local Government Debt* (Circular No.43), imposed tighter controls on new MCB issuance by clarifying that LGFV debts are not government debts and explicitly prohibits local governments from providing guarantees for LGFV debt. This policy differs from the 2017 personal responsibility rule in institutional design: whereas Circular No.43 strengthened administrative regulation of debt issuance, the 2017 rule linked default consequences more directly to officials' career outcomes through lifelong accountability. Table 8 compares the effects of these two policy shocks. After controlling for the interaction associated with Circular No.43, the post-2017 interaction remains negative and significant in the full sample, and its coefficient is larger in absolute value than that on the 2014 policy interaction. In the restricted sample spanning both policy changes, only the post-2017 interaction remains significant, while the 2014 interaction becomes insignificant. Taken together, these findings suggest that the baseline result reflects the distinct pricing relevance of the 2017 personal responsibility rule, above and beyond the broader regulatory

tightening introduced by Circular No.43.

Third, we replace primary-market issuance spreads with ChinaBond valuation spreads measured five trading days after listing. This alternative measure helps mitigate concerns that the baseline findings are driven by noise in issuance pricing, government-influenced buyers, or underwriter negotiations in the primary market. Compared with issuance spreads, ChinaBond valuation spreads more closely reflect market pricing based on public information. Table 9 shows that the core interaction remains significantly negative in both the full and restricted samples. The triple interaction with fiscal revenue also remains positive and significant, consistent with the baseline evidence on the fiscal channel. These results show that the main findings are not specific to primary-market issuance spreads and continue to hold when bond pricing is measured using ChinaBond valuation yields.

Overall, the baseline result is specific to party secretaries, distinct from the earlier 2014 debt-policy tightening, and robust to alternative spread measurement.

6 Conclusion

This paper develops a framework to study how political accountability shapes local debt markets. We model the MCB market as an incomplete-information bargaining game between local officials and investors and show that the 2017 personal responsibility rule makes observable official characteristics more informative about bailout incentives. In particular, remaining time to retirement becomes a credible signal of officials' incentives to prevent bond distress. Guided by this framework, our empirical analysis focuses on separating officials' incentive effects from their inherent ability. We find that after 2017, stronger career incentives are associated with lower issuance spreads and greater bond issuance, indicating that investors rationally price officials' career concerns into expected implicit support. More broadly, the findings qualify the conventional view that stronger accountability necessarily disciplines local borrowing: when markets interpret account-

ability as strengthening ex post bailout incentives, the policy may weaken rather than reinforce debt-control objectives.

Beyond the effect of political accountability on local government debt, the analysis also yields two broader insights into the functioning of the MCB market. First, in addition to the standard risk premium, the persistent positive yield spreads of MCBs appear to reflect an investment-dependence premium: local governments, facing growth targets and limited bargaining power, must offer higher yields to obtain financing. Second, the market pricing of heterogeneous official characteristics related to bailout incentives contributes to cross-city dispersion in MCB spreads, particularly for short-maturity bonds. In this sense, the personal responsibility rule does not generate more uniform borrowing behavior across cities. Instead, by making political incentives more salient to investors, it can increase heterogeneity in financing conditions.

From a policy perspective, our findings suggest that raising officials' personal default costs alone may be insufficient to curb local debt accumulation. Once investors interpret the personal responsibility rule as strengthening officials' incentives to prevent bond distress, officials with stronger career concerns may obtain cheaper financing rather than face tighter market discipline. A policy intended to harden budget constraints can therefore, through endogenous market pricing, partially restore expectations of implicit support. Debt-control policies, therefore, need to reduce the incentives for debt-financed growth directly, rather than relying primarily on ex post penalties tied to default outcomes. This could involve weakening the linkage between short-run economic performance and political advancement and clarifying the boundary between government obligations and corporate liabilities. More broadly, the results show that the effectiveness of accountability reforms depends not only on enforcement but also on how they reshape market expectations.

Declaration of AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT5.2 developed by OpenAI in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Table 1. Descriptive Statistics of MCBs, Officials, and Cities

Variable	Unit	Full Sample					Restricted Sample						
		Obs.	Mean	S.D.	p25	Median	p75	Obs.	Mean	S.D.	p25	Median	p75
Panel A: Municipal Corporate Bond Characteristics													
Coupon	bps	24,474	540.4	153.8	420.0	539.0	658.0	9,467	521.0	143.4	415.0	513.0	629.0
Spread	bps	17,873	207.9	113.7	125.0	193.9	276.0	6,736	191.8	113.1	113.8	175.3	258.6
Val. yield	bps	11,106	561.3	132.5	463.0	559.4	661.2	4,997	549.0	125.3	457.3	547.1	640.1
Val. spread	bps	9,547	206.3	98.08	133.3	194.3	264.2	4,081	198.6	94.77	130.4	184.9	252.2
Size	¥100m	24,474	8.866	5.914	5.000	8.000	10.00	9,467	8.821	5.775	5.000	8.000	10.00
Maturity	years	24,474	4.159	2.592	2.000	5.000	7.000	9,467	3.845	2.509	1.000	4.000	5.000
Panel B: Official Characteristics (City-Month Level)													
TTR	years	51,804	7.048	3.451	4.500	6.583	9.167	15,074	6.406	3.117	4.167	6.083	8.250
Pre-mayor	0/1	37,914	0.699					10,316	0.700				
Central exp.	0/1	51,798	0.129					15,074	0.155				
Mayor TTR	years	51,605	8.935	3.598	6.250	8.667	11.33	15,065	7.891	3.398	5.333	7.583	10.17
Panel C: City Characteristics (City-Year Level Economic Variables & City-Month Level Land Transfer Variables)													
GDP	¥100m	3,705	2,070	2,561	698.6	1,253	2,377						
GDP pc	¥10k	3,705	4.709	3.262	2.540	3.814	5.897						
Fiscal rev.	%	3,705	7.341	2.645	5.479	6.953	8.851						
Land rev.	¥100m	44,457	10.46	28.98	0.728	2.794	8.906	12,525	12.87	30.03	0.984	3.779	11.63
Land rev. pc	¥	44,457	210.9	517.3	25.62	80.15	220.2	12,525	245.4	423.3	35.50	104.6	272.3

Notes: The full sample consists of 24,474 MCBs issued by LGFVs in prefecture-level and sub-provincial cities from 2008 to 2020. The restricted sample includes only bonds issued under party secretaries whose tenure spans the July 2017 personal responsibility rule. Variation in observations across Panel A variables reflects incomplete matching with risk-free benchmark rates and partial coverage in the ChinaBond database. *TTR* (Time to Retirement) measures the years remaining until statutory retirement age for party secretaries (60 years for prefecture-level officials; 63 years for sub-provincial officials); *Mayor TTR* is defined analogously for mayors. *Pre-mayor* equals 1 if the party secretary was previously mayor of the same city. *Central exp.* equals 1 if the party secretary has prior central government working experience. Panel B contains more observations than Panel A because official characteristics are matched at the city-month level; missing values arise during leadership transitions. *Fiscal rev.* denotes the ratio of general public budget revenue to GDP (in percent) and serves as an inverse proxy for municipal fiscal stress. Restricted-sample statistics are not reported for city-year variables in Panel C because these variables are not linked to specific official tenures.

Table 2. Time to Retirement, Fiscal Stress, and MCB Yield Spreads

Dependent variable: <i>Spread</i>	Full sample		Restricted sample	
	(1)	(2)	(3)	(4)
<i>Post</i>	36.74*** (7.765)	138.8*** (25.87)	19.98*** (7.139)	144.0*** (30.25)
<i>TTR</i>	23.62*** (3.394)	25.16*** (4.175)	-2.186 (2.299)	-3.501 (5.627)
<i>TTR</i> $\times$ <i>Post</i>	-3.231*** (0.954)	-13.13*** (3.345)	-2.548*** (0.877)	-15.10*** (3.656)
<i>Fiscal rev.</i>		0.945 (2.386)		-1.079 (4.337)
<i>Fiscal rev.</i> $\times$ <i>Post</i>		-12.23*** (2.917)		-15.11*** (3.402)
<i>TTR</i> $\times$ <i>Fiscal rev.</i>		-0.173 (0.275)		0.0307 (0.547)
<i>TTR</i> $\times$ <i>Fiscal rev.</i> $\times$ <i>Post</i>		1.185*** (0.378)		1.538*** (0.412)
<i>Maturity</i>	0.184 (0.382)	0.187 (0.382)	-1.317** (0.526)	-1.305** (0.524)
$\log(\text{GDP})$	-136.3*** (26.19)	-142.5*** (26.64)	-203.4*** (37.52)	-186.4*** (39.14)
$\log(\text{GDP per capita})$	46.05*** (11.89)	41.18*** (11.91)	74.42*** (22.54)	49.45** (22.75)
Mean Dep. Var.	201.3	201.3	194.4	194.4
Year FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	No	No
Observations	11401	11401	6533	6533
Adjusted R-squared	0.342	0.343	0.338	0.342

Notes: *Spread* is the MCB issuance yield minus the matching risk-free rate (in basis points). *Post* is an indicator for bonds issued after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age. *Fiscal rev.* is the ratio of general public budget revenue to local GDP (in percent). Columns (1)–(2) use the full sample of MCBs issued during 2008–2020, and Columns (3)–(4) restrict the sample to bonds issued under party secretaries whose tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Time to Retirement, Fiscal Stress, and MCB Issuance Volume

Dependent variable: $\log(Size)$	Full sample		Restricted sample	
	(1)	(2)	(3)	(4)
<i>Post</i>	-0.102* (0.0557)	-0.551*** (0.161)	-0.152*** (0.0529)	-0.759*** (0.194)
<i>TTR</i>	-0.00261 (0.0186)	-0.0412* (0.0236)	-0.0784*** (0.0165)	-0.126*** (0.0320)
<i>TTR</i> $\times$ <i>Post</i>	0.0199*** (0.00657)	0.0707*** (0.0218)	0.0200*** (0.00618)	0.0795*** (0.0245)
<i>Fiscal rev.</i>		-0.0361** (0.0149)		-0.0736*** (0.0247)
<i>Fiscal rev.</i> $\times$ <i>Post</i>		0.0519*** (0.0182)		0.0676*** (0.0218)
<i>TTR</i> $\times$ <i>Fiscal rev.</i>		0.00432** (0.00170)		0.00594* (0.00308)
<i>TTR</i> $\times$ <i>Fiscal rev.</i> $\times$ <i>Post</i>		-0.00587** (0.00247)		-0.00650** (0.00277)
<i>Maturity</i>	0.0639*** (0.00243)	0.0640*** (0.00243)	0.0454*** (0.00292)	0.0457*** (0.00293)
$\log(GDP)$	0.402** (0.164)	0.436*** (0.166)	0.534** (0.233)	0.447* (0.239)
$\log(GDP \text{ per capita})$	0.100 (0.0955)	0.101 (0.0962)	-0.196 (0.163)	-0.192 (0.166)
Mean Dep. Var.	2.046	2.046	1.982	1.982
Year FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	No	No
Observations	14212	14212	9255	9255
Adjusted R-squared	0.199	0.199	0.158	0.159

Notes: *Size* is the issuance volume of MCBs (in 100 million yuan). *Post* is an indicator for bonds issued after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age. *Fiscal rev.* is the ratio of general public budget revenue to local GDP (in percent). Columns (1)–(2) use the full sample of MCBs issued during 2008–2020, and Columns (3)–(4) restrict the sample to bonds issued under party secretaries whose tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4. Time to Retirement and Land Transfer Revenue

Dependent variable:	Full sample		Restricted sample	
	(1) <i>Land Rev.</i>	(2) <i>Land Rev. per capita</i>	(3) <i>Land Rev.</i>	(4) <i>Land Rev. per capita</i>
<i>Post</i>	-0.273 (1.278)	-21.37 (20.90)	0.317 (1.391)	2.023 (21.25)
<i>TTR</i>	-1.982 (1.292)	-67.02*** (13.37)	-2.415** (1.005)	-75.48*** (13.47)
<i>TTR</i> $\times$ <i>Post</i>	0.560** (0.224)	11.14*** (3.102)	0.436** (0.188)	6.819** (2.765)
<i>log(GDP)</i>	3.618* (1.997)	27.64 (50.63)	26.56*** (5.926)	256.3*** (69.82)
<i>log(GDP per capita)</i>	-1.617 (1.358)	-35.36 (26.38)	-17.06*** (5.627)	-156.0** (63.79)
Mean Dep. Var.	9.239	192.7	12.87	245.4
Year FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	No	No
Observations	37596	37596	12525	12525
Adjusted R-squared	0.330	0.182	0.485	0.364

Notes: *Land Rev.* and *Land Rev. per capita* denote the local government's monthly land transfer revenue (in 100 million yuan) and land transfer revenue per capita (in yuan), respectively. *Post* is an indicator for months after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age. Columns (1)–(2) use the full city-month sample, and Columns (3)–(4) restrict the sample to city-month observations under party secretaries whose tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Personal Responsibility Rule and Cross-City Yield Dispersion

Dependent variable: <i>S.D.</i>	1-year bond		5-year bond	
	(1) Coupon	(2) Spread	(3) Coupon	(4) Spread
<i>Post</i>	21.31** (9.456)	20.10** (9.911)	-1.776 (10.83)	-2.116 (11.01)
<i>time</i>	-0.994** (0.400)	-1.025** (0.400)	1.007** (0.381)	1.048*** (0.387)
<i>Post</i> $\times$ <i>time</i>	2.682*** (0.538)	2.873*** (0.588)	-0.584 (0.437)	-0.577 (0.453)
<i>Constant</i>	38.20*** (5.925)	37.01*** (5.929)	115.7*** (9.099)	115.8*** (9.148)
Observations	72	70	73	71
Adjusted R-squared	0.437	0.428	0.312	0.315

Notes: *S.D.* is the monthly cross-city standard deviation of MCB coupon rates or yield spreads (in basis points). *Post* is an indicator for months after July 2017. *Time* measures the number of months relative to the 2017 policy implementation and is normalized to zero in the event month. The sample consists of monthly city-level dispersion measures constructed from MCBs issued during 2008–2020. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Heterogeneity in Officials' Career Background

Dependent variable: <i>Spread</i>	Full sample				Restricted sample			
	Mayor-turned		Central experience		Mayor-turned		Central experience	
	(1) Yes	(2) No	(3) Yes	(4) No	(5) Yes	(6) No	(7) Yes	(8) No
<i>Post</i>	39.54*** (9.987)	32.55 (20.69)	27.20 (21.73)	42.77*** (8.298)	23.12** (9.701)	-2.177 (19.47)	-2.844 (17.97)	28.93*** (7.881)
<i>TTR</i>	23.26*** (4.398)	23.96*** (6.854)	17.72** (7.307)	24.97*** (3.827)	-3.393 (6.427)	9.381 (10.39)	-2.923 (4.907)	-1.444 (2.860)
<i>TTR</i> × <i>Post</i>	-3.686*** (1.196)	-2.197 (2.625)	-2.742 (2.409)	-3.881*** (1.043)	-2.881** (1.149)	0.593 (2.505)	-0.884 (2.115)	-3.480*** (0.981)
<i>Maturity</i>	-0.565 (0.517)	0.980 (0.706)	-0.0561 (0.909)	0.262 (0.421)	-1.010 (0.777)	-2.357*** (0.859)	-2.829*** (0.977)	-0.675 (0.622)
<i>log(GDP)</i>	-196.5*** (33.20)	-75.02 (55.53)	-253.3*** (65.09)	-123.7*** (30.19)	-277.0*** (60.06)	-185.7*** (70.26)	-19.58 (120.8)	-211.1*** (40.06)
<i>log(GDP per capita)</i>	47.98*** (16.74)	32.82* (18.54)	44.40** (19.59)	50.64*** (15.82)	76.88* (43.11)	63.35** (32.12)	-96.83 (95.85)	92.38*** (24.42)
Mean Dep. Var.	199.2	192.4	190.1	204.5	194.4	174.9	180.7	199.1
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	Yes	Yes	No	No	No	No
Observations	6870	2633	2506	8895	3896	1349	1658	4875
Adjusted R-squared	0.329	0.388	0.286	0.357	0.302	0.464	0.313	0.344

Notes: *Spread* is the MCB issuance yield minus the matching risk-free rate (in basis points). *Post* is an indicator for bonds issued after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age. Heterogeneity is examined by whether the party secretary was promoted from mayor of the same city (Columns (1)–(2) and (5)–(6)) and by central-government experience (Columns (3)–(4) and (7)–(8)). Columns (1)–(4) use the full sample of MCBs issued during 2008–2020, and Columns (5)–(8) restrict the sample to bonds issued under party secretaries whose tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7. Robustness: Mayoral vs. Secretarial Incentives

Dependent variable: <i>Spread</i>	Full sample			Restricted sample		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Post</i>	36.74*** (7.765)	14.84* (8.117)	40.62*** (9.829)	19.98*** (7.139)	8.180 (7.692)	20.46* (10.80)
<i>TTR</i>	23.62*** (3.394)		2.501*** (0.629)	-2.186 (2.299)		-8.922** (3.934)
<i>TTR</i> $\times$ <i>Post</i>	-3.231*** (0.954)		-4.233*** (0.881)	-2.548*** (0.877)		-1.844* (1.045)
<i>Mayor TTR</i>		21.19*** (3.368)	20.57*** (3.372)		13.24*** (3.414)	8.625 (5.645)
<i>Mayor TTR</i> $\times$ <i>Post</i>		-0.733 (0.911)	-0.216 (0.919)		0.0531 (0.856)	-0.685 (0.974)
<i>Maturity</i>	0.184 (0.382)	0.184 (0.379)	0.193 (0.379)	-1.317** (0.526)	-1.950*** (0.524)	-1.880*** (0.629)
<i>log(GDP)</i>	-136.3*** (26.19)	-150.7*** (31.38)	-145.7*** (31.41)	-203.4*** (37.52)	-180.6*** (35.15)	-159.3*** (51.76)
<i>log(GDP per capita)</i>	46.05*** (11.89)	47.25*** (11.24)	48.09*** (11.27)	74.42*** (22.54)	54.56*** (14.90)	80.97** (32.38)
Mean Dep. Var.	201.3	202.9	202.9	194.4	191.9	192.7
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Secretary FE	Yes	-	No	No	-	No
Mayor FE	-	Yes	Yes	-	No	No
Observations	11401	11839	11839	6533	6416	4361
Adjusted R-squared	0.342	0.361	0.362	0.338	0.366	0.369

Notes: *Spread* is the MCB issuance yield minus the matching risk-free rate (in basis points). *Post* is an indicator for bonds issued after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age, and *Mayor TTR* is defined analogously for the city mayor. Columns (1)–(3) use the full sample of MCBs issued during 2008–2020. Column (3) includes both secretary and mayor retirement measures but controls only for mayor fixed effects, in order to avoid potential collinearity. Column (4) uses the secretary-based restricted subsample, Column (5) the mayor-based restricted subsample, and Column (6) the subsample in which both tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. Robustness: 2017 vs. 2014 Policy

Dependent variable: <i>Spread</i>	Full sample			Restricted sample		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>TTR</i>	23.62*** (3.394)	20.35*** (3.344)	23.56*** (3.536)	-2.186 (2.299)	6.582*** (1.710)	-1.224 (10.32)
<i>Post</i>	36.74*** (7.765)		34.18*** (7.811)	19.98*** (7.139)		18.98 (14.99)
<i>TTR</i> $\times$ <i>Post</i>	-3.231*** (0.954)		-2.984*** (0.956)	-2.548*** (0.877)		-3.045* (1.842)
<i>Post-2014</i>		7.763 (8.932)	7.458 (8.932)		-2.076 (8.766)	19.34 (35.28)
<i>TTR</i> $\times$ <i>Post-2014</i>		-2.223*** (0.712)	-1.996*** (0.713)		-1.860*** (0.719)	0.295 (2.224)
<i>Maturity</i>	0.184 (0.382)	0.231 (0.382)	0.194 (0.382)	-1.317** (0.526)	-0.0603 (0.529)	-4.066*** (1.066)
<i>log(GDP)</i>	-136.3*** (26.19)	-135.1*** (26.15)	-135.8*** (26.17)	-203.4*** (37.52)	-122.6*** (35.03)	-22.22 (62.88)
<i>log(GDP per capita)</i>	46.05*** (11.89)	47.40*** (11.93)	46.89*** (11.95)	74.42*** (22.54)	45.55*** (14.87)	13.39 (32.75)
Mean Dep. Var.	201.3	201.3	201.3	194.4	202.5	208.4
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	Yes	No	No	No
Observations	11401	11401	11401	6533	5340	1379
Adjusted R-squared	0.342	0.341	0.342	0.338	0.276	0.330

Notes: *Spread* is the MCB issuance yield minus the matching risk-free rate (in basis points). *TTR* denotes the years until the local party secretary reaches the statutory retirement age. *Post* is an indicator for bonds issued after July 2017, and *Post-2014* is an indicator for bonds issued after September 2014. Columns (1)–(3) use the full sample of MCBs issued during 2008–2020. Column (4) restricts the sample to bonds issued under party secretaries whose tenures span the 2017 policy change, Column (5) to those whose tenures span the September 2014 policy change, and Column (6) to those whose tenures span both policy changes. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Robustness: ChinaBond Valuation Spreads

Dependent variable: <i>Val. Spread</i>	Full sample		Restricted sample	
	(1)	(2)	(3)	(4)
<i>Post</i>	28.50*** (8.353)	159.8*** (27.05)	23.52*** (7.534)	124.0*** (31.62)
<i>TTR</i>	11.08*** (4.225)	10.48** (5.336)	-3.870 (2.691)	-14.42** (7.283)
<i>TTR</i> $\times$ <i>Post</i>	-2.725*** (1.024)	-16.48*** (3.562)	-2.908*** (0.924)	-13.32*** (3.854)
<i>Fiscal rev.</i>		-0.795 (2.727)		-8.819* (4.766)
<i>Fiscal rev.</i> $\times$ <i>Post</i>		-16.11*** (3.131)		-12.89*** (3.600)
<i>TTR</i> $\times$ <i>Fiscal rev.</i>		0.0591 (0.355)		1.004 (0.709)
<i>TTR</i> $\times$ <i>Fiscal rev.</i> $\times$ <i>Post</i>		1.682*** (0.405)		1.348*** (0.435)
<i>Maturity</i>	1.413*** (0.505)	1.439*** (0.506)	-0.166 (0.668)	-0.137 (0.666)
<i>log(GDP)</i>	-143.4*** (33.35)	-151.1*** (33.26)	-182.4*** (42.74)	-165.6*** (45.03)
<i>log(GDP per capita)</i>	64.01*** (16.64)	56.66*** (16.75)	85.56*** (26.02)	61.41** (26.83)
Mean Dep. Var.	194.2	194.2	196.9	196.9
Year FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Secretary FE	Yes	Yes	No	No
Observations	6014	6014	3990	3990
Adjusted R-squared	0.346	0.349	0.337	0.344

Notes: *Val. Spread* is the ChinaBond valuation yield of MCB minus the matching risk-free rate (in basis points). *Post* is an indicator for bonds issued after July 2017. *TTR* denotes the years until the local party secretary reaches the statutory retirement age. *Fiscal rev.* is the ratio of general public budget revenue to local GDP (in percent). Columns (1)–(2) use the full sample of MCBs issued during 2008–2020, and Columns (3)–(4) restrict the sample to bonds issued under party secretaries whose tenures span the 2017 policy change. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

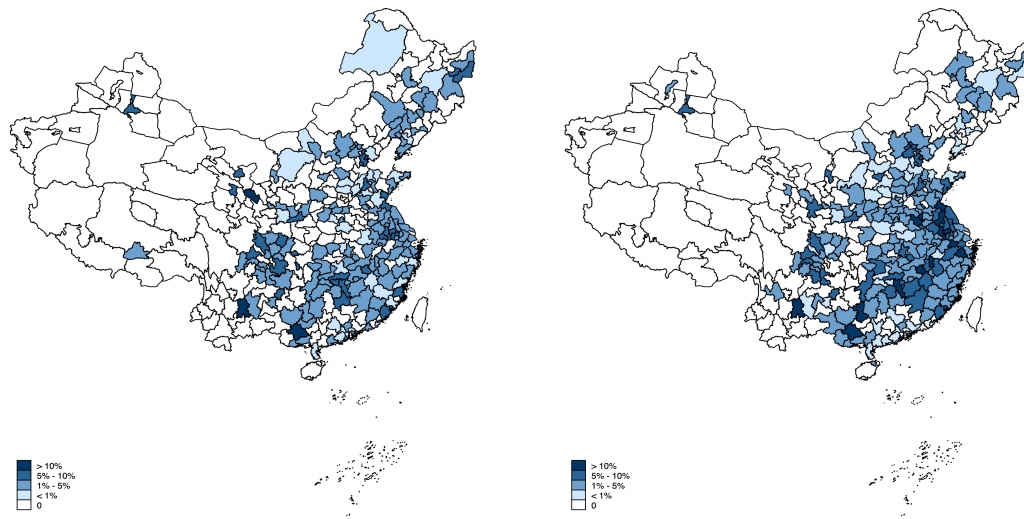


Figure 2. MCB issuance volumes in 2015 & 2020

Notes: The maps measure the ratio of annual MCB issuance volume to city GDP. Sample covers all prefecture-level administrative units in China, excluding Hong Kong, Macao, and Taiwan.

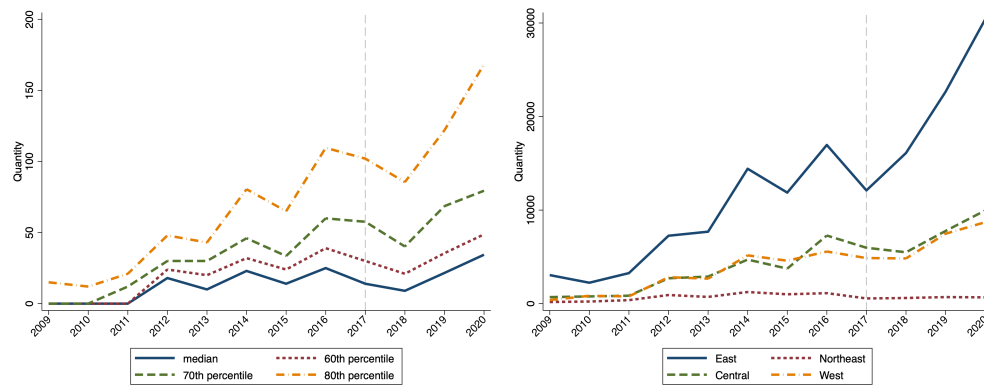


Figure 3. MCB issuance volume at prefectural and regional level

Notes: median/60th/70th/80th percentile refer to city-level issuance volume percentiles across all prefectures. Eastern/Central/Western/Northeast China follow China's official regional division standards.

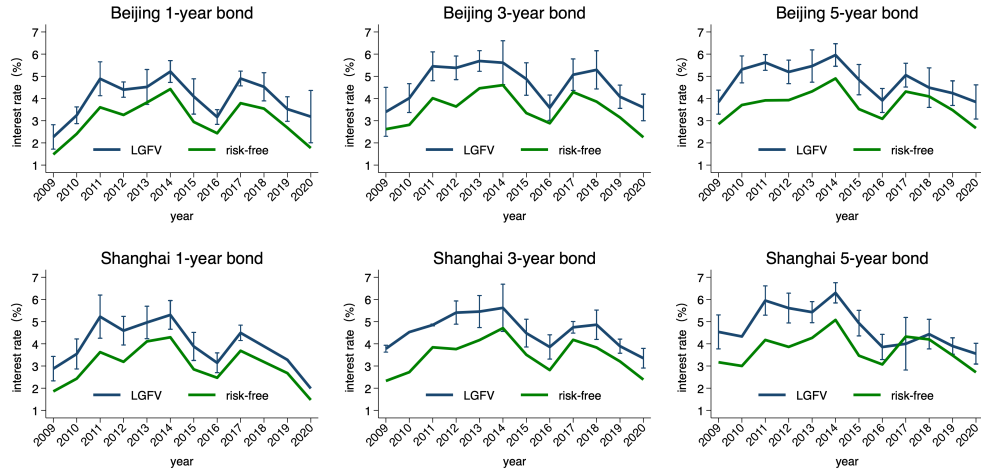


Figure 4. MCB issuance interest rates in Beijing and Shanghai

Notes: Error bars represent one standard deviation above and below the mean of MCB rates. Maturities (1/3/5-year) are matched to corresponding risk-free rate tenors.

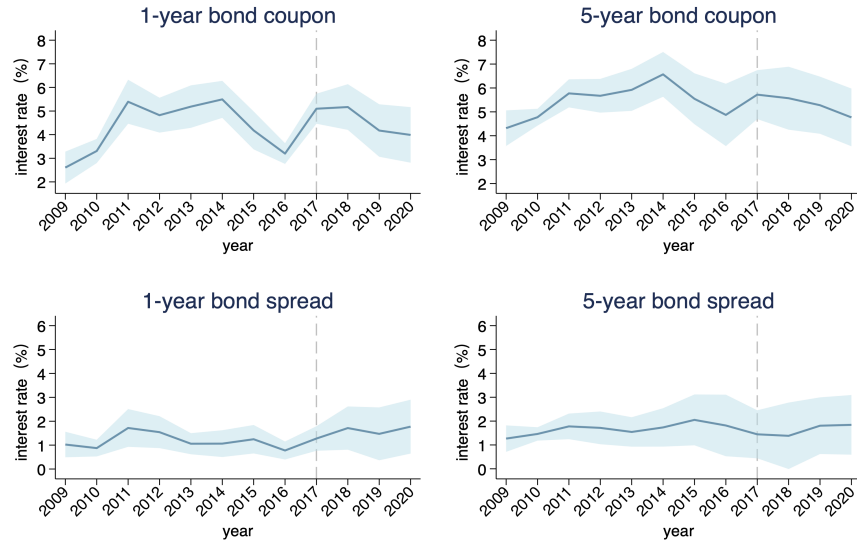


Figure 5. MCB issuance interest rates at prefectural level

Notes: The solid line denotes the volume-weighted mean across cities in each year, and the shaded area indicates the one-standard-deviation band of the city-level distribution.

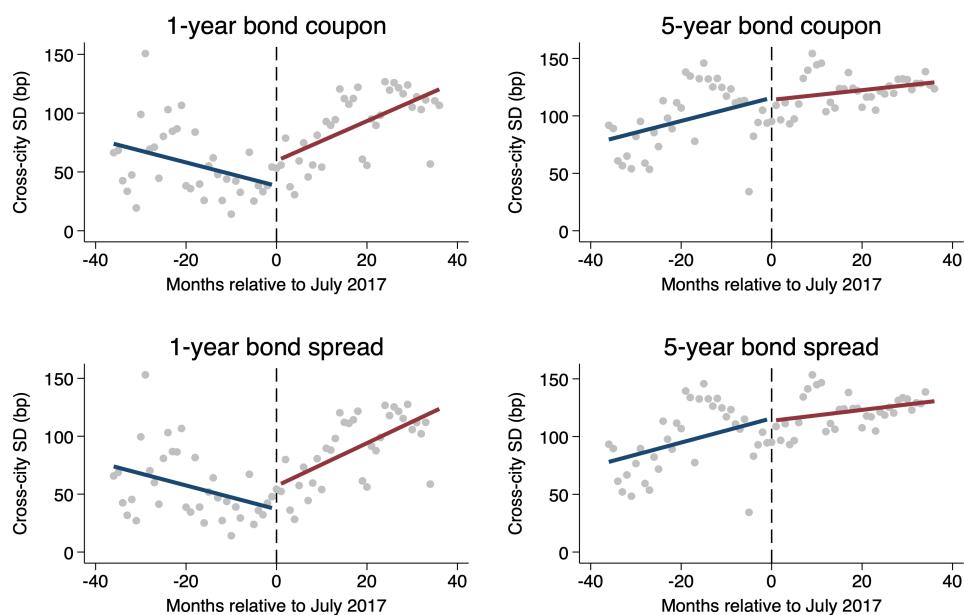


Figure 6. Regression-discontinuity style plot of cross-city dispersion of MCBs

Notes: Each dot represents the monthly cross-city standard deviation of MCB yields, and the fitted lines trace the estimated local trends before and after July 2017.

A Proof of the Propositions

In order to obtain testable predictions, we make the following assumption.

Assumption 1. *After the introduction of the personal responsibility rule, the expected probability that a local official bails out a distressed MCB is higher than that of central government intervention in all cities. Mathematically, this can be expressed as*

$$\forall z, \quad \lambda < 1 - F(c - \zeta(z)). \quad (1)$$

This assumption guarantees the uniqueness of comparative statics by establishing an upper bound for the probability of central intervention. Intuitively, the central government should refrain from excessive intervention in the repayment process of local debts and instead place greater reliance on the initiatives of local officials themselves. Without this assumption, the correlation between MCBs and the characteristics of local officials would be diminished. Local officials might then readily opt against making extra efforts to bail out fiscal stress, shifting such responsibilities to the central government, a scenario that is clearly undesirable for the central authority. Therefore, the central government will set λ at a relatively small value, a measure that aligns with the original rationale of this assumption.

A.1 Proof of Proposition 1

The partial differentiation of (15) yields

$$y''(x^*) \frac{\partial x^*(z, p)}{\partial z} = -p \left[\frac{(1 - \lambda + r_f) F'(c - \zeta(z)) \zeta'(z)}{(1 - pF(c - \zeta(z)))^2} - \hat{c}'(z) \right]. \quad (2)$$

Applying the Leibniz rule for differentiation under the integral sign to equation (10), we obtain

$$\hat{c}'(z) = F(c - \zeta(z)) \zeta'(z). \quad (3)$$

Substituting the last expression into (2), it follows that

$$\frac{\partial x^*(z, p)}{\partial z} = \frac{-p}{y''(x^*)} \left[\frac{(1 - \lambda + r_f)F'(c - \zeta(z))}{(1 - pF(c - \zeta(z)))^2} - F(c - \zeta(z)) \right] \zeta'(z). \quad (4)$$

The sign of the derivative in the last line depends on the term in the brackets. From the assumption that ε is uniformly distributed over $[0, 1]$, it follows that $F'(\varepsilon) = 1$. Assumption 1 implies

$$\frac{(1 - \lambda + r_f)F'(c - \zeta(z))}{(1 - pF(c - \zeta(z)))^2} - F(c - \zeta(z)) > 1 - \lambda + r_f - F(c - \zeta(z)) > r_f. \quad (5)$$

Since the production function satisfies $y''(x^*) < 0$, we establish that $\frac{\partial x^*(z, p)}{\partial z} > 0$.

Differentiating equation (16) with respect to z yields

$$\begin{aligned} \frac{\partial r^*(z, p)}{\partial z} = & -p \left[\frac{(1 - \lambda + r_f)\theta F'(c - \zeta(z))}{(1 - pF(c - \zeta(z)))^2} + (1 - \theta)F(c - \zeta(z)) \right] \zeta'(z) \\ & + (1 - \theta) \left(\frac{y(x)}{x} \right)' \Big|_{x^*} \cdot \frac{\partial x^*(z, p)}{\partial z}. \end{aligned} \quad (6)$$

As both terms on the right side of the equation are negative, it is proven that $\frac{\partial r^*(z, p)}{\partial z} < 0$.

A.2 Proof of Proposition 2

It follows from equation (4) that

$$\frac{\partial^2 x^*(z, p)}{\partial z \partial p} = \frac{-1}{y''(x^*)} \left[\frac{(1 - \lambda + r_f)F'(c - \zeta(z))(1 + pF(c - \zeta(z)))}{(1 - pF(c - \zeta(z)))^3} - F(c - \zeta(z)) \right] \zeta'(z). \quad (7)$$

Similarly to equation (5), Assumption 1 also implies that $\frac{\partial^2 x^*(z, p)}{\partial z \partial p} > 0$. It follows from equation (6) that

$$\begin{aligned} \frac{\partial^2 r^*(z, p)}{\partial z \partial p} = & - \left[\frac{(1 - \lambda + r_f)\theta F'(c - \zeta(z))(1 + pF(c - \zeta(z)))}{(1 - pF(c - \zeta(z)))^3} + (1 - \theta)F(c - \zeta(z)) \right] \zeta'(z) \\ & + (1 - \theta) \left(\frac{y(x)}{x} \right)' \Big|_{x^*} \cdot \frac{\partial^2 x^*(z, p)}{\partial z \partial p}. \end{aligned} \quad (8)$$

As both terms on the right side of the equation are negative, it is proved that $\frac{\partial^2 r^*(z,p)}{\partial z \partial p} < 0$. Thus, the negative effect of local official's characteristic on bond yield and the positive effect on bond issuance volume are amplified in cities with high fiscal risks.

A.3 Proof of Proposition 4

Let $\Omega_c = \{p_c, y_c(\cdot)\}$ denote the set of economic fundamentals for city c , where p_c is the probability of financial stress and $y_c(\cdot)$ represents the local production technology. Let z_c denote the personal characteristic of the official in city c . We define the variability of interest rates as the cross-sectional variance, denoted by $\text{Var}(r^*)$. According to the Law of Total Variance, the variance of the equilibrium interest rate can be decomposed into the variation explained by city fundamentals (between-group variance) and the residual variation due to other factors (within-group variance):

$$\text{Var}(r^*) = \text{Var}_\Omega (E[r^* \mid \Omega]) + E_\Omega [\text{Var}(r^* \mid \Omega)]. \quad (9)$$

Before the personal responsibility rule, the bargaining solution is characterized by equations (13) and (14). In this regime, the equilibrium interest rate r_{pre}^* is determined solely by city fundamentals Ω_c and global parameters (such as r_f and λ). Specifically, since the local official's bailout effort is not tied to observable personal characteristics z , investors treat officials as homogeneous conditional on city fundamentals. Thus, for any two cities i and j with identical fundamentals $\Omega_i = \Omega_j$, the equilibrium rates must be identical: $r_{pre,i}^* = r_{pre,j}^*$. Mathematically, this implies that the conditional variance is zero:

$$\text{Var}(r_{pre}^* \mid \Omega) = 0. \quad (10)$$

Substituting this into equation (9), the total variance is driven entirely by the heterogeneity in fundamentals: $\text{Var}(r_{pre}^*) = \text{Var}_\Omega(r_{pre}^*)$.

After the personal responsibility rule, the equilibrium is now characterized by equations (15) and (16). The interest rate r_{post}^* becomes a function of both the city fundamen-

tals Ω_c and the official's characteristic z_c . From Proposition 1, we have established that $\partial r^*/\partial z < 0$, indicating a strict monotonic relationship between the interest rate and the official's incentives. Consequently, even for cities with identical fundamentals ($\Omega_i = \Omega_j$), variations in officials' characteristics ($z_i \neq z_j$) will generate dispersion in interest rates. Assuming a non-degenerate distribution of z across officials, the conditional variance becomes strictly positive:

$$\text{Var}(r_{post}^* \mid \Omega) > 0. \quad (11)$$

Comparing the two regimes via equation (9), the introduction of the personal responsibility rule adds a positive term $E_\Omega[\text{Var}(r_{post}^* \mid \Omega)]$ to the variance decomposition. Assuming that the distribution of city fundamentals Ω remains stable and that specific official characteristics z are not systematically assigned to perfectly offset fundamental risks (i.e., z acts as an independent source of variation), the total variability of interest rates strictly increases:

$$\text{Var}(r_{post}^*) > \text{Var}(r_{pre}^*). \quad (12)$$